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1  %-----
2  % Two-Period Consumption Problem
3  %-----
4
5  clear
6  clc
7
8  %Printing the outputs as a text file if ==1
9  output_file=0;
10
11 %Problem Set-up
12 beta=0.9;
13 fun=@(x) -(log(x(1)) + beta*log(x(2)));
14
15 R0=1.05;
16 a_rhs=1;
17 A=[1 1/R0];
18 b=[a_rhs];
19 Aeq=[];
20 beq=[];
21 lb=[0; 0];
22 ub=[];
23 nonlcon = [];
24 x0=[0.5; 0.5];
25
26 %fmincon
27 [x,fval,exitflag,output,lambda,grad,hessian] = fmincon(fun,x0,A,b,Aeq,beq,lb,ub,nonlcon);
28
29 if output_file==1
30     dfile="two_periods_consumption_fmincon.txt";
31     if exist(dfile, 'file')
32         delete(dfile);
33     end
34     diary(dfile)
35     diary on
36 end
37
38 %Output
39 fprintf('beta=%.3f, R0=%.3f, A=%.3f\n',beta,R0,a_rhs);
40 fprintf('Optimal consumption in two periods: %.3f,%.3f\n',x(1),x(2));
41 fprintf('Lifetime Utility: %.5f\n',-fval);
42 fprintf('Multiplier (intertemporal budget const): %.5f\n',lambda.ineqlin);
43 fprintf('Multiplier (non-negativity const): %.5f, %.5f\n',lambda.lower(1),lambda.lower(2));
44
45 if output_file==1
46     diary off
47 end
48
49 %Plot
50 x1_vec=linspace(0, a_rhs, 100);
51 x2_vec=linspace(0, R0*a_rhs, 100);
52 [x1_mesh, x2_mesh] = meshgrid(x1_vec, x2_vec);
53 figure();
54 hold on;
55 grid on;
56 plot(x1_vec, R0*(a_rhs-x1_vec),'k', 'LineWidth', 1); %budget constraint
57 contour(x1_mesh, x2_mesh, log(x1_mesh)+beta*log(x2_mesh), [(-fval), (- fval)], 'LineWidth', 1,
    'LineColor', 'b');

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58 contour(x1_mesh, x2_mesh, log(x1_mesh)+beta*log(x2_mesh), [1.25*(-fval), 0.75*(-fval)],
'LineWidth', 1, 'LineColor', 'k', 'LineStyle', '--');
59 plot(x(1),x(2),'ro','LineWidth', 1);
60 xlabel('$x_{1}$','Interpreter','latex','FontSize',12);
61 ylabel('$x_{2}$','Interpreter','latex','FontSize',12);
62 title('Two-Period Consumption Problem','Interpreter','latex','FontSize',14);
63 xlim([0,a_rhs]);
64 ylim([0,R0*a_rhs]);
65

```